AvePoint®

# sTMS Interface Functional Specification

## Temasek Polytechnic – VDI System

Version 1.0

AvePoint

# Document Version

| Version | Effective Date | Summary of Changes | Created/ Updated By |
|---|---|---|---|
| 0.1 | 04/03/2022 | Initial version of TP VDI IFS | AvePoint |
| 0.2 | 31/03/2022 | Revised based on feedback from TP. 2.2 Add file retention policy, change the frequency and timing description of input interface file and output interface file; Update 3.2 Output Interface Fields: subject action description; 4.1 add assumption 6 for subject process logic | AvePoint |
| 0.3 | 19/04/2022 | Revised based on feedback from TP<br>Update 2.1 System Flow diagram and 2.2 Interface Design Summary<br>• Change interface file extension to csv.pgp<br>Update 3.2 Output Interface Fields<br>• Change field name School to School of Student in the Student section<br>• Change field name School to School of Student in the Lecturer section<br>• Add a field name called "School of Subject" in the Subject section<br>Update 3.3 VDI_STUDENT, VDI_LECTURER, VDI_SUBJECT_DETAIL sample files | AvePoint |
| 0.4 | 27/04/2022 | Revised based on feedback from TP<br>2.1 System Flow diagram and 2.2 Interface Design Summary<br>• Update interface file name<br>• Update system flow diagram<br>Document Acceptance Form<br>• Update Approved By to TP PIT lead<br>3.2 Output Interface Fields<br>• Add a remark for "School of Lecturer" field | AvePoint |
| 1.0 | 05/05/2022 | Official version for acceptance and sign-off | AvePoint |

AvePoint

# Document Acceptance Form

**Prepared By:**

| | |
|---|---|
| Martin Hong | Signature: DocuSigned by: Martin Hong 254C335124C5452... |
| Business Analyst, AvePoint | Date: May 9, 2022 \| 01:52 EDT |

**Review By:**

| | |
|---|---|
| Mint Lang | Signature: DocuSigned by: Mint Lang C7CEC25CE540453... |
| Solution Architect, AvePoint | Date: May 8, 2022 \| 21:46 EDT |
| Murray Qiao | Signature: DocuSigned by: Murray Qiao 10D34FF1345C49F... |
| Technical Manager, AvePoint | Date: 2022年5月5日 \| 04:51 EDT |

**Approved By:**

| | |
|---|---|
| Sng Choon Leng | Signature: DocuSigned by: Choon Leng Sng F3EBE7337B9B482... |
| Director, Academic Affairs, Temasek Polytechnic | Date: May 8, 2022 \| 23:42 EDT |

# Table of Contents

# 1. Introduction

## 1.1 Purpose of this document

Temasek Polytechnic VDI system (Online Virtual Lab Environment) is a platform that provides a virtual lab environment and necessary online capabilities to serve based on subject needs. It grants access for students enrolled in this subject to utilize the virtual lab facilities as well as the lecturer who is running and overseeing this subject. On top of this, the VDI system enables provision scalability based on subject schedules.

This Interface Functional Specification (IFS) aims to provide the agreed baseline of communication requirement and detailed technical specification between the SaaS shared Training Management System (sTMS) and Temasek Polytechnic VDI system (Online Virtual Lab Environment) for student, lecturer, and subject information.

## 1.2 Intended Audience

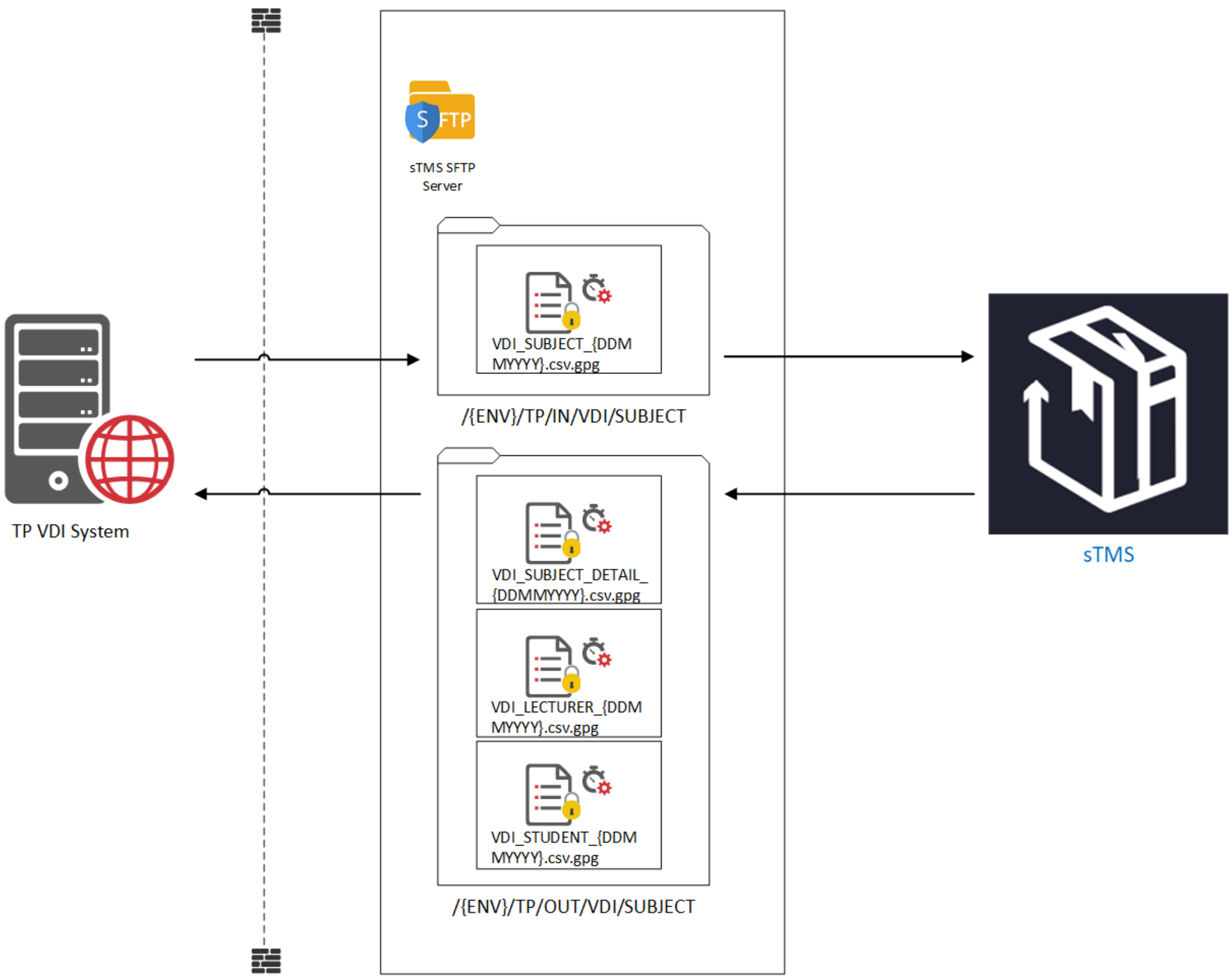
This technical document is used for system-to-system communication. Therefore, the intended audiences are the technical stakeholders of both systems.

# 2. Technical Overview

There are 3 roles involved in the interface diagram.

- **TP VDI System**: Temasek Polytechnic VDI system.
- **sTMS SFTP Server**: A SFTP server hosted in Azure that stores CSV files used for data exchange.
- **sTMS**: shared Training Management System.

## 2.1 System Flow



| SN | Description |
|---|---|
| 1 | TP VDI system generates a *subject list file* and pushes it to the sTMS SFTP server after encryption using the PGP protocol. |
| 2 | sTMS grabs the *subject list file* on a pre-defined schedule. |
| 3 | sTMS configures Timer jobs to generate *subject detail list, lecturer list, student list files,* and pushes them to sTMS SFTP server after encryption with PGP protocol. |
| 4 | TP VDI system pulls the *subject detail list, lecturer list,* and *student list files* on demand. |

## 2.2 Interface Design Summary

**Input Interface File**

| | |
|---|---|
| Interface File Name | VDI_SUBJECT_{DDMMYYYY}.csv.gpg |
| Data Source System | Temasek Polytechnic VDI System |
| Data Destination System | shared Training Management System |
| Mode of Interface | CSV file |
| Method | SFTP |
| File Path | {ENV}/TP/IN/VDI/SUBJECT<br>ENV: UAT, PROD |
| Encryption Method | PGP |
| Frequency | Once or several times per semester (configurable) |
| Timing | One week before the start date of the semester and the subsequent 4 weeks (configurable) |
| File Retention Policy | Files are retained for a maximum of 30 days in the SFTP server, after this period the files will be automatically deleted. |
| Possible Exception and Exception Handling | 1. File loss or transfer failure due to network exception.<br>Exception handling steps:<br>• sTMS support team and VDI system admin shall be informed by email of the transfer failure.<br>• AP team checks the relevant logs to troubleshoot the cause of the failure.<br>• If the issue is due to the transfer file could not be found in SFTP:<br>    ○ AP informs VDI system admin to re-deposit the file in SFTP, and AP manually reruns the timer job.<br><br>2. Data error.<br>Exception handling steps: |

| | |
|---|---|
| | • sTMS support team and VDI system admin shall be informed by email of the transfer failure.<br>• AP team checks the relevant logs and files to troubleshoot the cause of the failure (data error types are listed below).<br>  ○ Field exceeds max length allowed<br>  ○ Field is mandatory but missing<br>  ○ Field is corrupted<br>• AP informs VDI system admin to re-deposit the correct file in SFTP and AP manually reruns the timer job. |

**Output Interface File**

| | |
|---|---|
| Interface File Name | VDI_STUDENT_{DDMMYYYY}.csv.gpg<br>VDI_LECTURER_{DDMMYYYY}.csv.gpg<br>VDI_SUBJECT_DETAIL_{DDMMYYYY}.csv.gpg |
| Data Source System | shared Training Management System |
| Data Destination System | Temasek Polytechnic VDI System |
| Mode of Interface | CSV file |
| Method | SFTP |
| File Path | {ENV}/TP/OUT/VDI/SUBJECT<br>ENV: UAT, PROD |
| Encryption Method | PGP |
| Data Syncing Mode | First-time full data sync + Delta data sync for subsequent days (sTMS will not send empty files if there is no delta data) |
| Frequency | Daily job for 4 weeks (configurable) |
| Timing | After the input file from TP VDI system is synchronized to sTMS, schedule the job to start one week before the starting date of the semester (configurable) |
| File Retention Policy | Files are retained for a maximum of 30 days in the SFTP server, after this period the files will be automatically deleted. |
| Possible Exception and Exception Handling | 1. File loss or transfer failure due to network exception.<br>Exception handling steps:<br>• sTMS support team shall be informed by email of the transfer failure.<br>• AP team checks the relevant logs to troubleshoot the cause of the failure.<br>• If the issue is due to the transfer files could not be found in SFTP:<br>  ○ AP redeposits the files in SFTP.<br><br>2. Data error. |

| | |
|---|---|
| | Exception handling steps:<br>• sTMS support team shall be informed by email of the transfer failure.<br>• AP team checks the relevant logs and files to troubleshoot the cause of the failure (data error types are listed below).<br>○ Field exceeds max length allowed<br>○ Field is mandatory but missing<br>○ Field is corrupted<br>• AP redeposits the correct files in SFTP. |

## 2.3 Business Flow

Flow to generate Student Interface File

| SN | Description |
|---|---|
| 1 | For a given subject, the system locates the corresponding TP course intake daily |
| 2 | sTMS checks the student course intake enrollment status daily |
| 3 | sTMS identifies the changes of students (delta changes) for the given course intake during the date of checking<br>Delta changes include:<br>1. New students enroll under course intake and start the given subject<br>2. Students withdraw or defer courses resulting in their removal from the given subject |
| 4 | sTMS consolidates the delta changes, generates the output interface file, and deposits it accordingly. |

Flow to generate Lecturer Interface File

| SN | Description |
|---|---|
| 1 | For a given subject, the system checks the full list of sessions daily |
| 2 | sTMS summarizes all lecturers for the given subject based on the latest full list of sessions, compares with the existing lecturer list associated with the given subject, identifies the changes of lecturers (delta changes).<br>Delta changes include:<br>1. New lecturer is assigned to the given subject |

| | |
|---|---|
| | 2. Existing lecturer is removed from the given subject |
| 3 | sTMS consolidates the delta changes of the lecturers, generates the lecturer output interface file, and deposits it accordingly. |

### Flow to generate Subject Interface Files

| SN | Description |
|---|---|
| 1 | For a given subject, the system checks the session and updates the status daily |
| 2 | sTMS identifies the changes of sessions (delta changes) for the given subject (newly created, updated, deleted) during the date of checking<br><br>Delta changes include<br>1. New session is created for the given subject<br>2. Existing session is removed from the given subject<br>3. The information or attribute of a session is updated, e.g. No of student changes, timing changes, duration changes. |
| 3 | sTMS consolidates the delta changes of the sessions, generates the subject output interface file, and deposits it accordingly. |

AvePoint

# 3. Interface Fields

## 3.1 Input Interface Parameter

| Name | Description | Data Type | Length | Remarks |
|---|---|---|---|---|
| *Subject Code | Code of subject/module for which VDI account set up to be performed | Char | 10 | |

*The asterisk * represents the field is required.*

## 3.2 Output Interface Fields

| Name | Description | Data Type | Length | Remarks |
|---|---|---|---|---|
| **Student** | | | | |
| *Action | The indication of delta changes of the given student record | String | 1 | Refer **Action List** in Appendix for further information. |
| *Course Type | The type of course where the given subject is under | String | 10 | Sample values: CET |
| *Student Admission No | Student identification stored in the VDI system | String | 10 | Common ID from UAM |
| *Subject Code | Code to subject | String | 10 | |
| *Subject Name | Name of subject | String | 1~255 | |
| Lecture Group | Lecture group student attends | String | 0~255 | Leave blank as it is not in sTMS |
| Tutorial Group | Tutorial group student attends | String | 0~10 | Leave blank as it is not in sTMS |
| Practical Group | Practical group student attends | String | 0~10 | Leave blank as it is not in sTMS |
| *Course Code | Code to identify the course | String | 10 | |
| *Course Name | Name of course | String | 1~255 | |
| *School of Student | Code of school where the student belongs to | String | 0~10 | |
| **Lecturer** | | | | |
| *Action | The indication of delta changes of the given Lecturer record | String | 1 | Refer **Action List** in Appendix for further information. |
| *Login Id | Lecturer login ID stored in AAD | String | 50 | |
| *Lecturer Name | Name of lecturer | String | 1~100 | |
| *Subject Code | Code to subject | String | 10 | |

| | | | | |
|---|---|---|---|---|
| *Subject Name | Name of subject | String | 1~255 | |
| *Department of Lecturer | Code of department where the lecturer belongs to | String | 0~10 | E.g. BUS, TSA, IR, HR, etc. |
| **Subject** | | | | |
| *Action | The indication of delta changes of the given Subject record | String | 1 | Refer **Action List** in Appendix for further information. |
| *Subject Code | Code to subject | String | 10 | |
| *Subject Name | Name of subject | String | 1~255 | |
| *School of Subject | Code of school where the subject belongs to | String | 20 | |
| *Class Type | Options of how the class is conducted | String | 10 | Refer **Class Type List** in Appendix for further information. |
| Class Group | Class group name | String | 10 | |
| No of Students | Number of students attending the given subject | Integer | | |
| *Week Day | The day of the week when the subject is held | String | 10 | Sample values: Monday |
| *Time From | Starting time of the class | String | 5 | Format: hh: mm |
| *Time To | End time of the class | String | 5 | Format: hh: mm |
| *Weeks | Course period in terms of week | String | 10 | Sample value: 1-8 |
| *Length | Course duration in hours | Integer | 10 | Unit: hour |
| No of Staffs | Number of staff running this subject | Integer | 10 | |

*The asterisk * represents the field is required.*

## 3.3 Sample file

Below are the sample files provided by AP.

VDI_SUBJECT_03032021.csv VDI_STUDENT_03032021.csv VDI_LECTURER_03032021.csv VDI_SUBJECT_DETAIL_03032021.csv

# 4. Assumption and Constraints

## 4.1 Assumption

| No. | Assumptions |
|---|---|
| 1 | As sTMS shall be the core system for all the subject timetable-related information, any unmapped fields shall be left blank. |
| 2 | Subject in VDI is equivalent to module in sTMS. |
| 3 | The student shall be identified by UAM Common ID |
| 4 | The lecturer shall be identified by AAD Login ID |
| 5 | The 3 outbound interface files can be sent out sequentially |
| 6 | sTMS only sends the full data of the subjects once. It means that if some of the same subjects are included in a new CSV file sent later in the same semester, these subjects shall not be sent the full data again. |
| | |

# 5. Appendix

## Action List

| CODE | DESCRIPTION |
|---|---|
| C | Create new record |
| U | Update existing record |
| D | Delete existing record |

## Class Type List

| CODE | DESCRIPTION |
|---|---|
| FTF | Face to face |